Date	31 October 2026
Team ID	XXXXXX
Project Name	Global Fuel Prices and Taxation Analysis (2020–2026)
Maximum Marks	2 Marks

Project Design Phase-II
Technology Stack (Architecture & Stack)

Define Your Technology Stack:
Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	**Programming / Query Language**	Language used to query, filter, transform, aggregate, and analyze fuel market data.	Tableau Calculated Fields & Level of Detail (LOD) Expressions	Used to compute complex metrics such as Tax Share %, Net Subsidy Burden, Regional Volatility Index, and Brent crude pass-through correlations across 84 countries.
2.	**Database / Data Warehouse**	Storage layer used to manage structured historical energy data for analytics and reporting.	Excel / CSV Extract Dataset (Global Fuel Database 2020–2026)	Houses structured fuel records (27,468 rows across Petrol, Diesel, LPG, tax structures, and Brent crude) and connects directly to Tableau without requiring a heavy database server.

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3.	**Data Ingestion**	Used to load data from source files into the analytical visualization environment.	Tableau Desktop Data Engine & Hyper Extract Connector	Loads multi-fuel pricing datasets directly into Tableau memory, enabling instant data refresh whenever underlying CSV/Excel files update.
4.	**Data Transformation and Business Logic**	Tools/Logic used to clean, join, filter, aggregate fuel data and implement fiscal metrics.	Tableau Data Source Tab, Calculated Fields, and Aggregation Functions	Handles currency normalization, geographic role assignments (Countries/Regions), and business logic for tax heatmaps, subsidy categorizations, and income level segmentations.
5.	**Data Visualization / Analytics**	Used to create interactive dashboards, visual analytics, trendlines, and scatter plots.	Tableau Desktop: Filled Maps, Heatmaps, Dual-Axis Line Charts, Scatter Plots, and Cross-Tabs	Builds interactive dashboards covering tax policy maps, diesel price logistics distributions, crude correlation models, and multi-fuel price trends over time (2020–2026).
6.	**Front-End / User Interface**	Interface through which stakeholders interact with dashboards, filters, and stories.	Tableau Interactive Dashboards shared via Tableau Public / Packaged Workbook (.twbx)	Provides policy consultants, logistics directors, and researchers with dynamic filters, tooltips, and parameter controls accessible directly via web browsers.